EQUIVARIANT GAMMA EXPANSION OF MATROID CHOW RINGS

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Editorial note. Original author statement, entire paired proof, definitions and supporting arguments retained. No added mathematical hypothesis, proof repair or correctness certification.

Editorial note. Full article context is retained to preserve internal references, prerequisite constructions, and native diagrams. Only this task primary statement is counted.

Editorial note. Mechanical package adaptation: source line 19 loads shuffle.sty, but the article contains no shuffle command. Only this unused package-loading line is excluded; mathematical macros and body retained.

Allowed standard prerequisites (editorial).

- Standard graph, matroid and enumerative combinatorics. The source article definitions, cited standard results and bibliography are retained.. Original prerequisites; no new assumptions or author-proof replacements.

1. Introduction

Given a finite sequence, there are many features that one can ask whether the sequence possesses. A sequence $\{a_i\}_{i=0}^d$ or a polynomial $f(t) = \sum_{i=0}^d a_i t^i$ of degree d is said to be *palindromic* if

$$a_j = a_{d-j} \quad \text{for all } j = 0, 1, \ldots, d.$$

It is said to be *unimodal* if there is some $0 \leq j \leq d$ such that

$$a_0 \leq a_1 \leq \cdots \leq a_j \geq \cdots \geq a_{d-1} \geq a_d.$$

A palindromic polynomial $f(t)$ can be expressed uniquely as

$$f(t) = \sum_{k=0}^{\lfloor \frac{d}{2} \rfloor} \gamma_k t^k (1+t)^{d-2k}.$$

The above expression is known as the γ-expansion of f and $(\gamma_1, \gamma_2, \ldots, \gamma_{\lfloor \frac{d}{2} \rfloor})$ is called the γ-vector of f. We say the polynomial f is *γ-positive* (or *γ-nonnegative*) if $\gamma_k \geq 0$ for all k. It is not hard to see that a polynomial being γ-positive implies that it is both palindromic and unimodal.

There are several reasons why the γ-positivity of a polynomial has attracted significant attention. One is the abundance of well-studied combinatorial sequences that are γ-positive; in many cases, their γ-coefficients admit interesting combinatorial interpretations, and sometimes this positivity phenomenon together with these interpretations can be lifted to a richer algebraic or representation-theoretic setting. A classical example is the γ-positivity of the Eulerian polynomial, first studied by Foata and Strehl [15], which was later extended to the level of symmetric functions [28, 29]. On the other hand, γ-positivity also appears naturally in geometric contexts. In particular, Gal conjectured that the h-polynomial of every flag simplicial sphere is γ-positive [16, Conjecture 2.1.7]. In this framework, the Eulerian polynomial arises as the h-polynomial of the dual permutahedron, which is itself a flag simplicial sphere. For more details about γ-positivity, we refer the readers to the survey [3] by Athanasiadis.

Let M be a loopless matroid on the ground set E. The *Chow ring of the matroid M* encodes the structure of the lattice of flats $\mathcal{L}(M)$ and is defined by

$$A(M) \coloneqq \mathbb{R}[x_F : F \in \mathcal{L}(M) \setminus \{\emptyset\}]/(I + J)$$

where $I = (x_F x_{F'} : F, F' \text{ not comparable})$ and $J = (\sum_{F: i \in x_F} x_F : i \in E)$. Similarly, the *augmented Chow ring of M* is defined by

$$\widetilde{A}(M) \coloneqq \mathbb{R}[\{x_F : F \in \mathcal{L}(M) \setminus \{[n]\}\} \cup \{y_i : i \in E\}]/(\widetilde{I} + \widetilde{J})$$

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where $\widetilde{I} = (x_F x_{F'} : F, F' \text{ not comparable}) + (y_i x_F : i \notin F)$ and $\widetilde{J} = (y_i - \sum_{F: i \notin F} x_F : i \in E)$. The ordinary Chow ring $A(M)$ can be recovered from the augmented one by quotienting out the ideal generated by the variables y_i. Despite the seemingly complicated presentation, the Chow ring $A(M)$ has a nice $\mathbb{R}$-basis, which we refered to as the *Feichtner-Yuzvinsky basis*

$$\mathrm{FY}(M) \coloneqq \left\{ x_{F_1}^{a_1} x_{F_2}^{a_2} \dots x_{F_\ell}^{a_\ell} : \begin{smallmatrix} \emptyset = F_0 \subsetneq F_1 \subsetneq F_2 \subsetneq \dots \subsetneq F_\ell \text{ for } 0 \leq \ell \leq n \\ 1 \leq a_i \leq \mathrm{rk}_M(F_i) - \mathrm{rk}_M(F_{i-1}) - 1 \end{smallmatrix} \right\}.$$

It is the standard monomials obtained from the Gröbner basis of the ideal $I + J$ found by Feichtner and Yuzvinsky [11]. A similar Feichtner-Yuzvinsky basis for $\widetilde{A}(M)$,

$$\widetilde{\mathrm{FY}}(M) \coloneqq \left\{ x_{F_1}^{a_1} x_{F_2}^{a_2} \dots x_{F_\ell}^{a_\ell} : \begin{smallmatrix} \emptyset \subsetneq F_1 \subsetneq F_2 \subsetneq \dots \subsetneq F_\ell \text{ for } 0 \leq \ell \leq n \\ 1 \leq a_1 \leq \mathrm{rk}_M(F_1), \\ 1 \leq a_i \leq \mathrm{rk}_M(F_i) - \mathrm{rk}_M(F_{i-1}) - 1 \text{ for } i \geq 2 \end{smallmatrix} \right\},$$

was found by the author [22, 23] and independently by Chris Eur (see [27]). Using these bases, one can compute the Hilbert polynomials of the corresponding Chow rings and augmented Chow rings, which are now commonly referred to as the *Chow polynomial* and the *augmented Chow polynomials of a matroid*.

The Chow rings and augmented Chow rings of matroids play central roles in the resolution of the long-standing Rota-Welsh conjecture and Dowling-Wilson top-heavy conjecture, respectively. A key ingredient in both proofs is that these rings satisfy the so-called Kähler package (see [1] for Chow rings and [6] for augmented Chow rings). This package includes a Poincaré duality, which implies the Chow and augmented Chow polynomials are palindromic, and the Hard Lefschetz Theorem, which implies these polynomials are unimodal. The Chow and augmented Chow polynomials of a matroid were proved to be γ-positive by Ferroni, Matherne, Stevens, and Vecchi [12], and independently by Wang [12, p.33], using a recursive argument based on the semismall decomposition introduced in [7]. However, an explicit expression or a combinatorial interpretation of the γ_k coefficients remains unknown. Moreover, these polynomials were conjectured to satisfy an even stronger positivity phenomenon. This was formulated by Huh and Stevens [35, Conjecture 4.3.3], and independently by Ferroni and Schröter [14, Conjecture 8.18] (for Chow polynomials) as follows.

Conjecture 1.1. *For every loopless matroid M, the Chow polynomial and the augmented Chow polynomial of M have only real roots.*

Taking into account group actions on matroids, Angarone, Nathanson, and Reiner [2] extended several known and conjectural inequalities for Chow polynomials to the equivariant (representation-theoretic) and Burnside (permutation-representation) setting. In particular, they conjectured that the Chow ring $A(M)$, viewed as a graded $\mathbb{C}G$-modules, is equivariantly γ-positive (see Section 3 for the definition), and moreover equivariantly PF_∞ [2, Conjecture 4.4 (ii)], which is an equivariant analog of Conjecture 1.1.

This paper is the full version of the extended abstract [25] presented at FPSAC 2025. We prove the equivariant γ-positivity conjecture of Angarone, Nathanson, and Reiner. Our approach further provides an explicit interpretation of the γ-coefficients, which is new even in the non-equivariant setting. We expect that this interpretation will provide further insight into Conjecture 1.1 and its equivariant analog.

In Section 3, we establish the equivariant γ-positivity conjecture. In Section 4, we apply our γ-positivity formulas to uniform matroids. From the perspective of matroid Chow rings, our results extend the combinatorial interpretations of the Charney-Davis quantity for uniform matroids studied by Hameister, Rao, and Simposon [18], as well as for general matroids by Liao [24]. From the perspective of symmetric functions and representations of symmetric groups, our results generalize the Schur-γ-positivity of the Eulerian and binomial Eulerian quasisymmetric functions studied by Shareshian and Wachs [28, 29], Linusson, Shareshian, and Wachs [26], and Athanasiadis [3]. In Section 5, we apply results from Section 4 to address a problem raised by Athanasiadis [3, Problem 2.5] concerning a (p, q)-analog of the γ-expansion of the binomial Eulerian polynomial. In Section 6, we comment on applications of our equivariant γ-positivity results on braid matroids. In Section 7, we discuss recent developments related to this work that have appeared since the preliminary version was posted on `arXiv`.

In the course of preparing this paper, we learned that Stump [36] independently obtained an interpretation of the (non-equivariant) γ-coefficients for Chow and augmented Chow polynomials of matroids. His approach uses a specialization of the *Poincaré extended* **ab**-*index* introduced in [9], expressing the γ-coefficient as the number of maximal chains in $\mathcal{L}(M)$ with prescribed descent set under

an R-labeling. His results agree with ours, as a consequence of Stanley [31, Theorem 3.1]. In future work, we aim to further investigate the relationship between the Feichtner-Yuzvinsky basis and the Poincaré extended **ab**-index and study the equivariant analog of the extended **ab**-index.

2. Group actions on posets

Let P be a finite poset with a unique minimal element $\hat{0}$ and a unique maximal element $\hat{1}$. If P is graded of rank n with rank function $\mathrm{rk}_P : P \longrightarrow \{0, 1, \ldots, n\}$, then for $S \subseteq [n-1]$, the *rank selected subposet* of P is

$$P_S := \{x \in P \ : \ \mathrm{rk}_P(x) \in S\} \cup \{\hat{0}, \hat{1}\}.$$

Consider a group G of automorphisms of P acting on the poset P. The action of G preserves the rank of the elements in P; hence for any $S \subseteq [n-1]$, the group G permutes the maximal chains in P_S. Let us denote $\alpha_P(S)$ the permutation representation of G generated by the maximal chains in P_S. Consider the virtual representation

$$\beta_P(S) := \sum_{T \subseteq S} (-1)^{|S|-|T|} \alpha_P(T),$$

which by Möbius inversion, is equivalent to

$$\alpha_P(S) = \sum_{T \subseteq S} \beta_P(T). \tag{1}$$

Here $\alpha_P(S)$ and $\beta_P(S)$ can be viewed as the equivariant analogs of the flag f-vector and flag h-vector of the poset P, respectively. See [34, Section 3.13] for more details.

Theorem 2.1 (Stanley [32])**.** *If P is Cohen-Macaulay, then $\beta_P(S)$ is a genuine representation of G and*

$$\beta_P(S) \cong_G \tilde{H}_{|S|-1}(P_S).$$

Throughout this paper, the poset P we care about is always the lattice of flats of a matroid. It is well known that the lattice of flats of a matroid is always Cohen-Macaulay.

3. Equivariant γ-positivity

Let G be a finite group. For a $\mathbb{C}G$-module M, let $[M]$ be the isomorphism class of $\mathbb{C}G$-modules containing M.

Definition 3.1. The Grothendieck ring $R_{\mathbb{C}}(G)$ of $\mathbb{C}G$-modules is a free abelian group having the transversal of isomorphism classes of simple $\mathbb{C}G$-modules $\{[S_i], \ldots, [S_{cc(G)}]\}$ as a $\mathbb{Z}$-basis, where $cc(G)$ is the number of conjugacy classes of G, with the addition and multiplication relations given by

$$[S_i] + [S_j] := [S_i \oplus S_j] \quad \text{and} \quad [S_i] \cdot [S_j] := [S_i \otimes_{\mathbb{C}} S_j]$$

for $1 \leq i, j \leq cc(G)$ and extended linearly over $\mathbb{C}$. Now every element A in $R_{\mathbb{C}}(G)$ has a unique expression as $a = \sum_{i=1}^{cc(G)} a_i[S_i]$. We say A is a *genuine representation* of G if $a_i \geq 0$ for all i, denoted by $A \geq_{R_{\mathbb{C}}(G)} 0$.

Given a graded $\mathbb{C}G$-module $V = \bigoplus_i V_i$, define the *equivariant Hilbert series* of V to be the formal power series

$$\mathsf{Hilb}_G(V, t) = \sum_i [V_i] t^i \in R_{\mathbb{C}}(G)[[t]].$$

Definition 3.2. Let $V = \bigoplus_{i=0}^{d} V_i$ be a finite dimensional graded $\mathbb{C}G$-module. We say that V is *(G-)equivariant γ-positive* if its equivariant Hilbert series can be expressed as

$$\mathsf{Hilb}_G(V, t) = \sum_{i=0}^{d} [V_i] t^i = \sum_{k=0}^{\lfloor \frac{d}{2} \rfloor} \gamma_k t^k (1+t)^{d-2k}$$

and the uniquely defined coefficient $\gamma_k \in R_{\mathbb{C}}(G)$ is a class of a genuine representation of G over $\mathbb{C}$ for all k, i.e. $\gamma_k \geq_{R_{\mathbb{C}}(G)} 0$ for all k.

Let M be a loopless matroid of rank r on the ground set E with the lattice of flats $\mathcal{L}(M)$. Write $A(M)_{\mathbb{C}} \coloneqq A(M) \otimes_{\mathbb{R}} \mathbb{C}$ and $\widetilde{A}(M)_{\mathbb{C}} \coloneqq \widetilde{A}(M) \otimes_{\mathbb{R}} \mathbb{C}$.

From now on, let G be a group of automorphisms of the matroid M. The induced action of G on the Chow ring $A(M)_{\mathbb{C}} = \bigoplus_{i=0}^{r-1} A^i_{\mathbb{C}}$ endows each graded piece $A^i_{\mathbb{C}}$ a $\mathbb{C}G$-module structure. The same holds for the augmented Chow ring $\widetilde{A}(M)_{\mathbb{C}} = \bigoplus_{i=0}^{r} \widetilde{A}^i_{\mathbb{C}}$. The following conjecture was proposed by Angarone, Nathanson, and Reiner.

Conjecture 3.3 ([2, Conjecture 5.2])**.** *For any (loopless) matroid M of rank r and any group G of automorphisms of M, the matroid Chow ring $A(M)_{\mathbb{C}} = \bigoplus_{i=0}^{r-1} A^i_{\mathbb{C}}$ is (G-)equivariant γ-positive.*

In the remainder of this section, we prove Conjecture 3.3 in Theorem 3.5, as well as its augmented counterpart in Theorem 3.9. Note that the Feichtner-Yuzvinsky bases $\mathrm{FY}(M)$ and $\widetilde{\mathrm{FY}}(M)$ form permutation bases of $A(M)_{\mathbb{C}}$ and $\widetilde{A}(M)_{\mathbb{C}}$, respectively, under the action of G.

Proposition 3.4. *Let M be a loopless matroid of rank r.*

(i) *The G-equivariant Hilbert series of $A(M)_{\mathbb{C}}$ is given by*

$$\mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \phi_{S,r}(t)[\alpha_{\mathcal{L}(M)}(S)],$$

where for a subset $S = \{s_1 < s_2 < \cdots < s_\ell\} \subseteq [r-1]$,

$$\phi_{S,r}(t) \coloneqq t^{|S|}[s_1 - 1]_t[s_2 - s_1 - 1]_t \cdots [s_\ell - s_{\ell-1} - 1]_t[r - s_\ell]_t.$$

In particular, $\phi_{\emptyset,r}(t) = [r]_t$.

(ii) *The G-equivariant Hilbert series of $\widetilde{A}(M)_{\mathbb{C}}$ is given by*

$$\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \psi_{S,r}(t)[\alpha_{\mathcal{L}(M)}(S)],$$

where for a subset $S = \{s_1 < s_2 < \cdots < s_\ell\} \subseteq [r-1]$,

$$\psi_{S,r}(t) = \begin{cases} t^{|S|}[s_1]_t[s_2 - s_1 - 1]_t \ldots [s_\ell - s_{\ell-1} - 1]_t[r - s_\ell]_t & \text{if } S \neq \emptyset \\ [r+1]_t & \text{if } S = \emptyset \end{cases}.$$

Proof. For (i), recall that the Feichtenr-Yuzvinsky basis $\mathrm{FY}(M)$ consists of monomials $x_{F_1}^{a_1} x_{F_2}^{a_2} \ldots x_{F_\ell}^{a_\ell}$ indexed by chains (including the empty chain)

$$\emptyset \neq F_1 \subsetneq F_2 \subsetneq \cdots \subsetneq F_\ell \subseteq E$$

such that the exponent a_j satisfies $1 \leq a_j \leq \mathrm{rk}(F_j) - \mathrm{rk}(F_{j-1}) - 1$ for $j = 1, 2, \ldots, \ell$, where we set $F_0 = \emptyset$. For any subset $S = \{s_1 < s_2 < \cdots < s_\ell\} \subseteq [r-1]$, the permutation $\mathbb{C}G$-module generated by chains in $\mathcal{L}(M)$ with rank set S is

$$\begin{aligned} \alpha_{\mathcal{L}(M)}(S) &= \mathbb{C}G\{F_1 \subsetneq \cdots \subsetneq F_\ell : \mathrm{rk}(F_i) = s_i \ \forall i\}. \\ &\cong \mathbb{C}G\{F_1 \subsetneq \cdots \subsetneq F_\ell \subsetneq E : \mathrm{rk}(F_i) = s_i \ \forall i\}. \end{aligned}$$

Hence, as a graded $\mathbb{C}G$-module, $A(M)_{\mathbb{C}}$ decomposes as a direct sum of $\alpha_{\mathcal{L}(M)}(S)$ over all $S \subseteq [r-1]$.

Assume that $\mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \phi_{S,r}(t)[\alpha_{\mathcal{L}(M)}(S)]$ for some polynomials $\phi_{S,r}(t)$. To determine $\phi_{S,t}(t)$, we consider a map $f : \mathrm{FY}(M) \longrightarrow 2^{[r-1]}$ defined by

$$f(x_{F_1}^{a_1} x_{F_2}^{a_2} \ldots x_{F_\ell}^{a_\ell} x_E^i) = \{\mathrm{rk}(F_1), \mathrm{rk}(F_2), \ldots, \mathrm{rk}(F_\ell)\}$$

where all flats F_i are distinct and are not E. Since for any $S = \{s_1 < \cdots < s_\ell\} \subseteq [r-1]$, the inverse image of f on S consists of monomials

$$f^{-1}(S) = \left\{ x_{F_1}^{a_1} x_{F_2}^{a_2} \ldots x_{F_\ell}^{a_\ell} x_E^i : \begin{smallmatrix} \emptyset = F_0 \subsetneq F_1 \subsetneq \cdots \subsetneq F_\ell \subsetneq E \text{ with } \mathrm{rk}(F_j) = s_j \\ 1 \leq a_j \leq s_j - s_{j-1} - 1 \text{ for } j = 1, 2 \ldots, \ell \\ 0 \leq i \leq r - s_\ell - 1 \end{smallmatrix} \right\},$$

Therefore,

$$\begin{aligned} \phi_{S,r}(t) &= (t + t^2 + \cdots + t^{s_1 - 1}) \ldots (t + t^2 + \cdots + t^{s_\ell - s_{\ell-1} - 1})(1 + t + \cdots + t^{r - s_\ell - 1}) \\ &= t^{|S|}[s_1 - 1]_t[s_2 - s_1 - 1]_t \ldots [s_\ell - s_{\ell-1} - 1]_t[r - s_\ell]_t. \end{aligned}$$

In particular, when S is an empty set, the inverse image $f^{-1}(\emptyset) = \{x_E^i \ : \ 0 \leq i \leq r-1\}$ and $\phi_{\emptyset,r}(t) = [r]_t$.

For (ii), the argument is analogous. The basis $\widetilde{\mathrm{FY}}(M)$ consists of monomials indexed by chains as above, with exponents satisfying $1 \le a_1 \le \mathrm{rk}(F_1)$ and $1 \le a_j \le \mathrm{rk}(F_j) - \mathrm{rk}(F_{j-1}) - 1$ for $j \ge 2$. Thus, we write

$$\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \psi_{S,r}(t)\, [\alpha_{\mathcal{L}(M)}(S)].$$

Define $g : \widetilde{\mathrm{FY}}(M) \longrightarrow 2^{[r-1]}$ by

$$g(x_{F_1}^{a_1} \cdots x_{F_\ell}^{a_\ell} x_E^i) = \{\mathrm{rk}(F_1), \ldots, \mathrm{rk}(F_\ell)\}.$$

Then counting $g^{-1}(S)$ for $S = \{s_1 < \ldots < s_\ell\} \subseteq [r-1]$ gives

$$\psi_{S,r}(t) = \begin{cases} t^{|S|}[s_1]_t[s_2 - s_1 - 1]_t \cdots [s_\ell - s_{\ell-1} - 1]_t[r - s_\ell]_t & \text{if } S \neq \emptyset \\ [r+1]_t & \text{if } S = \emptyset \end{cases},$$

as claimed. □

The following theorem establishes Conjecture 1.1 and provides explicit expressions for the (equivariant) γ-coefficients of $A(M)$. It also generalizes the positivity of the (equivariant) Charney–Davis quantities for $A(M)$ obtained in [24, Lemma 4.9].

Theorem 3.5. *For any loopless matroid M of rank r on the ground set E with the lattice of flats $\mathcal{L}(M)$, the Chow ring $A(M)$ is equivariant γ-positive with the following γ-expansion*

$$\begin{aligned} \mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) &= \sum_{S \in \mathrm{Stab}([2,r-1])} [\beta_{\mathcal{L}(M)}(S)] t^{|S|} (1+t)^{r-1-2|S|} \\ &= \sum_{k=0}^{\lfloor \frac{r-1}{2} \rfloor} \left(\sum_{\substack{S \in \mathrm{Stab}([2,r-1]) \\ |S|=k}} [\beta_{\mathcal{L}(M)}(S)] \right) t^k (1+t)^{r-1-2k}, \end{aligned}$$

where $\beta_{\mathcal{L}(M)}(S) \cong \tilde{H}_{|S|-1}(\mathcal{L}(M)_S)$ as a $\mathbb{C}G$-module

Proof. Notice that the polynomial $\phi_{S,r}(t)$ in Proposition 3.4 (i) is equal to 0 if $s_j - s_{j-1} = 1$ for some $1 \le j \le \ell$. Hence $\phi_{S,r}(t)$ takes nonzero value only when $S \in \mathrm{Stab}([2, r-1])$. The identity in Proposition 3.4 can be rephrased as follows:

$$\begin{aligned} \mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) &= \sum_{S \in \mathrm{Stab}([2,r-1])} \phi_{S,r}(t) [\alpha_{\mathcal{L}(M)}(S)] \\ &= \sum_{S \in \mathrm{Stab}([2,r-1])} \phi_{S,r}(t) \sum_{T : T \subseteq S} [\beta_{\mathcal{L}(M)}(T)] \\ &= \sum_{T \in \mathrm{Stab}([2,r-1])} \left(\sum_{T \subseteq S \subseteq [r-1]} \phi_{S,r}(t) \right) [\beta_{\mathcal{L}(M)}(T)], \end{aligned}$$

where the second equality follows from equation (1). It turns out that this complicated sum has a neat reformulation:

$$\sum_{T \subseteq S \subseteq [r-1]} \phi_{S,r}(t) = t^{|T|}(1+t)^{r-1-2|T|}.$$

This fact and its proof will be presented in Lemma 3.6. Combining it with Theorem 2.1, which states that $\beta_{\mathcal{L}(M)}(T)$ is isomorphic to $\tilde{H}_{|T|-1}(\mathcal{L}(M)_T)$ as $\mathbb{C}G$-modules, we obtain

$$\begin{aligned} \mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) &= \sum_{T \in \mathrm{Stab}([2,r-1])} [\beta_{\mathcal{L}(M)}(T)] t^{|T|} (1+t)^{r-1-2|T|} \\ &= \sum_{T \in \mathrm{Stab}([2,r-1])} [\tilde{H}_{|T|-1}(\mathcal{L}(M)_T)] t^{|T|} (1+t)^{r-1-2|T|}. \end{aligned}$$

□

Remark 3.1. The γ-expansion in Theorem 3.5, together with its augmented counterpart in Theorem 3.9, recovers and extends several known γ-expansions from [28], [29], and [3]. Thus, the present approach yields a unified framework for these results; see Section 4 for further details.

Remark 3.2. It is relatively rare for equivariant γ-positivity to hold in such generality. To the best of the author's knowledge, another instance of this phenomenon appears in the context of *Rees products of posets* by Athanasiadis [4]. In that setting, a key step in establishing equivariant γ-positivity is [4, Lemma 3.1], which involves a pair of identities closely resembling Lemma 3.6 and its augmented counterpart, Lemma 3.10. It would be of interest to understand whether there is a common underlying structure that explains this parallel between the two settings.

Lemma 3.6. *For a positive integer $n \geq 2$ and a subset $T \in \mathrm{Stab}([2, n-1])$,*

$$\sum_{T \subseteq S \subseteq [n-1]} \phi_{S,n}(t) = t^{|T|}(1+t)^{n-1-2|T|}, \tag{2}$$

where $\phi_{S,n}(t) = 0$ if $S \notin \mathrm{Stab}([2, n-1])$ and for $S = \{s_1 < \cdots < s_\ell\} \in \mathrm{Stab}([2, n-1])$,

$$\phi_{S,n}(t) = t^{|S|}[s_1 - 1]_t[s_2 - s_1 - 1]_t \dots [s_\ell - s_{\ell-1} - 1]_t[n - s_\ell]_t.$$

We give a combinatorial proof of Lemma 3.6. This is done by double-counting a combinatorial model constructed below. The model is motivated by Angarone, Nathanson, and Reiner's combinatorial proof of [2, Theorem 1.1].

Definition 3.7. For $n \geq 2$, let $\mathcal{W}_n$ be the set of sequences $\mathbf{w} = w_1 w_2 \dots w_{n-1}$ over the alphabet set $\{\bullet, \times, _(\text{a blank space})\}$ satsifying the following conditions:

(W1) If $w_i = \bullet$, then $w_{i-1} = \times$ (hence $w_1 \neq \bullet$ and there is no consecutive $\bullet$ in $\mathbf{w}$).

Let $S = \{s_1 < s_2 < \dots < s_\ell\} \in \mathrm{Stab}([2, n-1])$ be the set of indices s with $w_s = \bullet$.

(W2) If there is $s_i < j < s_{i+1}$ such that $w_j = \times$, then $w_j = w_{j+1} = \dots = w_{s_{i+1}-1} = \times$.

(W3) If there is $s_\ell < j < n$ such that $w_j = \times$, then $w_j = w_{j+1} = \dots = w_{n-1} = \times$.

For $\mathbf{w} \in \mathcal{W}_n$, let $\mathsf{Dot}(\mathbf{w}) = \{i \in [n-1] : w_i = \bullet\}$ and $\mathsf{cro}(\mathbf{w})$ be the number of crossings $\times$ in $\mathbf{w}$.

Proof of Lemma 3.6. We prove (2) by double-counting

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})}.$$

First counting. Fix a subset $S \in \mathrm{Stab}([2, n-1])$, say $S = \{s_1 < \dots < s_\ell\}$. For a sequence $\mathbf{w} = w_1 \dots w_{n-1} \in \mathcal{W}_n$ with $\mathsf{Dot}(\mathbf{w}) = S$, we have $w_{s_j} = \bullet$ for all j. By (W1), each $\bullet$ forces $w_{s_j - 1} = \times$. Between consecutive positions s_{j-1} and s_j, the symbol at $s_j - 1$ is fixed to be $\times$, and we may choose any number of additional consecutive $\times$ immediately to its left, ranging from 0 to $s_j - s_{j-1} - 2$. This contributes a factor

$$t\left(1 + t + \cdots + t^{s_j - s_{j-1} - 2}\right) = t[s_j - s_{j-1} - 1]_t.$$

A similar argument applies to the initial segment before s_1 and the final segment after s_ℓ, with the latter contributing $[n - s_\ell]_t$. It follows that, if S is nonempty,

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) = S}} t^{\mathsf{cro}(\mathbf{w})} = t^{|S|}[s_1 - 1]_t[s_2 - s_1 - 1]_t \cdots [s_\ell - s_{\ell-1} - 1]_t[n - s_\ell]_t = \phi_{S,n}(t);$$

While $S = \emptyset$, $\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) = \emptyset}} t^{\mathsf{cro}(\mathbf{w})} = 1 + t + \cdots + t^{n-1} = [n]_t = \phi_{\emptyset,n}(t)$. Summing over all $S \supseteq T$ gives

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = \sum_{T \subseteq S \subseteq [n-1]} \phi_{S,n}(t).$$

Second counting. Fix $\mathbf{w} \in \mathcal{W}_n$ with $\mathsf{Dot}(\mathbf{w}) \supseteq T$. For each $i \in T$, we have $w_i = \bullet$, and hence by (W1), $w_{i-1} = \times$. These forced $\times$ contribute a factor $t^{|T|}$. The remaining positions, namely those not of the form i or $i-1$ for $i \in T$, form a set of size $n-1-2|T|$. On these positions, we may independently choose whether to place a $\times$ or leave a blank, contributing a factor $(1+t)^{n-1-2|T|}$. Given such a choice, the conditions (W2) and (W3) uniquely determine the placement of the remaining $\bullet$ symbols so that $\mathbf{w} \in \mathcal{W}_n$. Therefore,

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = t^{|T|}(1+t)^{n-1-2|T|}.$$

Comparing the two expressions completes the proof. □

In the following, we illustrate the procedure in the proof of Lemma 3.6 by some examples.

Example 3.8. Let $n = 12$ and $T = \{3, 9\} \in \mathrm{Stab}([2, 11])$.

First counting. Fix $S \supseteq T$.

Case 1: If $S = \{3, 5, 9\}$, we place $\bullet$ at positions $3, 5, 9$, and by (W1) we must place $\times$ at $2, 4, 8$. The remaining positions are initially blank:

1	2	3	4	5	6	7	8	9	10	11
_	$\times$	$\bullet$	$\times$	$\bullet$	_	_	$\times$	$\bullet$	_	_

Between consecutive $\bullet$'s, we may extend the forced $\times$'s to the left. For instance, between $w_5 = \bullet$ and $w_9 = \bullet$, we may choose

$$(_, _, \times), \quad (_, \times, \times), \quad (\times, \times, \times),$$

contributing $t + t^2 + t^3 = t[3]_t$. Proceeding similarly for all segments, we obtain

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_{12} \\ \mathsf{Dot}(\mathbf{w}) = \{3,5,9\}}} t^{\mathsf{cro}(\mathbf{w})} = t[2]_t \cdot t[1]_t \cdot t[3]_t \cdot [3]_t = t^3 [2]_t [1]_t [3]_t [3]_t.$$

Case 2: If $S = \{3, 9, 11\}$, we place $\bullet$ at $3, 9, 11$ and $\times$ at $2, 8, 10$:

1	2	3	4	5	6	7	8	9	10	11
_	$\times$	$\bullet$	_	_	_	_	$\times$	$\bullet$	$\times$	$\bullet$

Since $11 = n - 1 \in S$, there is no trailing segment after the last $\bullet$. The contributions from the segments give

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_{12} \\ \mathsf{Dot}(\mathbf{w}) = \{3,9,11\}}} t^{\mathsf{cro}(\mathbf{w})} = t[2]_t \cdot t[5]_t \cdot t[1]_t = t^3 [2]_t [5]_t [1]_t.$$

Second counting. We count sequences $\mathbf{w}$ with $\mathsf{Dot}(\mathbf{w}) \supseteq T = \{3, 9\}$. Then $w_3 = w_9 = \bullet$ and $w_2 = w_8 = \times$ are forced. The remaining positions

$$\{1, 4, 5, 6, 7, 10, 11\}$$

can be chosen independently to be either $\times$ or blank, contributing

$$t^{|\{2,8\}|}(1 + t)^{|\{1,4,5,6,7,10,11\}|} = t^2(1 + t)^{12-1-2\cdot 2}.$$

For example, if we choose $w_1 = w_4 = w_5 = w_{10} = \times$, we obtain

1	2	3	4	5	6	7	8	9	10	11
$\times$	$\times$	$\bullet$	$\times$	$\times$	_	_	$\times$	$\bullet$	$\times$	_

.

There is then a unique way to insert additional $\bullet$'s so that $\mathbf{w} \in \mathcal{W}_{12}$, namely:

1	2	3	4	5	6	7	8	9	10	11
$\times$	$\times$	$\bullet$	$\times$	$\times$	$\bullet$	_	$\times$	$\bullet$	$\times$	$\bullet$

.

This illustrates the identity

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_{12} \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = t^2(1 + t)^7.$$

The following theorem is the augmented counterpart of Theorem 3.5. It generalizes the positivity of the (equivariant) Charney–Davis quantities for the augmented Chow ring $\widetilde{A}(M)$ obtained in [24, Lemma 4.12]. The proof follows the same strategy as in the Chow case, with a modified combinatorial model.

Theorem 3.9. *For any loopless matroid M of rank r on the ground set E with the lattice of flats $\mathcal{L}(M)$, the augmented Chow ring $\widetilde{A}(M)_{\mathbb{C}}$ is equivariant γ-positive with the following γ-expansion*

$$\begin{aligned} \mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) &= \sum_{S \in \mathrm{Stab}([r-1])} [\beta_{\mathcal{L}(M)}(S)] t^{|S|} (1 + t)^{r-2|S|} \\ &= \sum_{k=0}^{\lfloor \frac{r}{2} \rfloor} \left(\sum_{\substack{S \in \mathrm{Stab}([r-1]) \\ |S| = k}} [\beta_{\mathcal{L}(M)}(S)] \right) t^k (1 + t)^{r-2k}. \end{aligned}$$

where $\beta_{\mathcal{L}(M)}(S) \cong \tilde{H}_{|S|-1}(\mathcal{L}(M)_S)$ as $\mathbb{C}G$-modules.

Proof. The argument is parallel to that of Theorem 3.5. By Proposition 3.4 (ii), we have

$$\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \psi_{S,r}(t)\, [\alpha_{\mathcal{L}(M)}(S)].$$

Since $\psi_{S,r}(t) = 0$ whenever S contains consecutive integers, the sum reduces to $S \in \mathrm{Stab}([r-1])$. Thus,

$$\begin{aligned}\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) &= \sum_{S \in \mathrm{Stab}([r-1])} \psi_{S,r}(t)[\alpha_{\mathcal{L}(M)}(S)] \\ &= \sum_{S \in \mathrm{Stab}([r-1])} \psi_{S,r}(t) \sum_{T \subseteq S} [\beta_{\mathcal{L}(M)}(T)] \\ &= \sum_{T \in \mathrm{Stab}([r-1])} \left(\sum_{T \subseteq S \subseteq [r-1]} \psi_{S,r}(t) \right) [\beta_{\mathcal{L}(M)}(T)].\end{aligned}$$

By Lemma 3.10 below, the inner sum satisfies

$$\sum_{T \subseteq S \subseteq [r-1]} \psi_{S,r}(t) = t^{|T|}(1+t)^{r-2|T|},$$

Combining this with Theorem 2.1, we obtain

$$\begin{aligned}\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) &= \sum_{T \in \mathrm{Stab}([r-1])} [\beta_{\mathcal{L}(M)}(T)] t^{|T|}(1+t)^{r-2|T|} \\ &= \sum_{T \in \mathrm{Stab}([r-1])} [\tilde{H}_{|T|-1}(\mathcal{L}(M)_T)] t^{|T|}(1+t)^{r-2|T|}.\end{aligned}$$

□

Lemma 3.10. *For $n \geq 2$ and subset $T \in \mathrm{Stab}([n-1])$,*

$$\sum_{T \subseteq S \subseteq [n-1]} \psi_{S,n}(t) = t^{|T|}(1+t)^{n-2|T|} \tag{3}$$

where $\psi_{S,n}(t) = 0$ if $S \notin \mathrm{Stab}([n-1])$ and for $S = \{s_1 < \cdots < s_\ell\} \in \mathrm{Stab}([n-1])$,

$$\psi_{S,n}(t) = \begin{cases} t^{|S|}[s_1]_t[s_2 - s_1 - 1]_t \dots [s_\ell - s_{\ell-1} - 1]_t[n - s_\ell]_t & \text{if } S \neq \emptyset \\ [n+1]_t & \text{if } S = \emptyset \end{cases}.$$

We use another combinatorial model slightly different from $\mathcal{W}_n$ in the non-augmented case to prove Lemma 3.10.

Definition 3.11. For $n \geq 2$, let $\widetilde{\mathcal{W}}_n$ be the set of sequences $\mathbf{w} = w_0 w_1 w_2 \dots w_{n-1}$ of length n over the alphabet set $\{\bullet, \times, _(\text{a blank space})\}$ satisfying conditions (W1), (W2), and (W3) from Definition 3.7.

Condition (W1) implies that $w_0 \neq \bullet$ and no two consecutive entries of $\mathbf{w}$ are both $\bullet$. As in the non-augmented case, we define $\mathsf{Dot}(\mathbf{w})$ to be the set of positions occupied by $\bullet$, and $\mathsf{cro}(\mathbf{w})$ to be the number of $\times$'s in $\mathbf{w}$ for $\mathbf{w} \in \widetilde{\mathcal{W}}_n$.

Proof of Lemma 3.10. The argument is parallel to that of Lemma 3.6. We compute

$$\sum_{\substack{\mathbf{w} \in \widetilde{\mathcal{W}}_n \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})}$$

in two ways.

First counting. Fix $S = \{s_1 < \cdots < s_\ell\} \in \mathrm{Stab}([n-1])$ with $S \supseteq T$. As in the non-augmented case, the structure of $\mathbf{w} \in \widetilde{\mathcal{W}}_n$ with $\mathsf{Dot}(\mathbf{w}) = S$ yields

$$\sum_{\substack{\mathbf{w} \in \widetilde{\mathcal{W}}_n \\ \mathsf{Dot}(\mathbf{w}) = S}} t^{\mathsf{cro}(\mathbf{w})} = \psi_{S,n}(t),$$

where the contribution from each interval $[s_{j-1}, s_j]$ is given by $[s_j - s_{j-1} - 1]_t$ except when $j = 1$ it is $[s_1]_t$, and the final segment contributes either $[n - s_\ell]_t$ or is absorbed into the special case $S = \emptyset$. Summing over all $S \supseteq T$ gives

$$\sum_{\substack{\mathbf{w}\in\widetilde{\mathcal{W}}_n \\ \mathsf{Dot}(\mathbf{w})\supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = \sum_{T\subseteq S\subseteq[n-1]} \psi_{S,n}(t).$$

Second counting. We instead construct sequences with $\mathsf{Dot}(\mathbf{w}) \supseteq T$ directly. First, for each $i \in T$, we force the pattern $w_i = \bullet$ and $w_{i-1} = \times$, contributing a factor $t^{|T|}$. After removing these forced positions, there remain $n - 2|T|$ free positions. In each such position, we independently choose either a $\times$ or a blank space, contributing a factor $(1+t)^{n-2|T|}$. Given any such choice, there is a unique way to insert additional $\times$'s so that the resulting sequence lies in $\widetilde{\mathcal{W}}_n$. Hence,

$$\sum_{\substack{\mathbf{w}\in\widetilde{\mathcal{W}}_n \\ \mathsf{Dot}(\mathbf{w})\supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = t^{|T|}(1+t)^{n-2|T|}.$$

Equating the two expressions completes the proof. □

Remark 3.3. In a forthcoming paper, Ferroni, Matherne, and Vecchi [13] generalize the Chow and augmented Chow polynomials of matroids (geometric lattice) to a more general class of posets. They prove these polynomials are γ-positive for Cohen–Macaulay posets, using a strategy analogous to that of the present paper, though without the combinatorial proofs of Lemma 3.6 and Lemma 3.10 (See Section 7).

4. Uniform and q-uniform matroids

In this section, we consider the special case where M is the uniform matroid $U_{r,n}$ of rank r and $G = \mathfrak{S}_n$ is the symmetric group. In this setting, the lattice of flats can be identified with a rank-selected subposet of the Boolean lattice $B_n = (2^{[n]}, \subseteq)$. The equivariant γ-coefficients $\tilde{H}_{|R|-1}((B_n)_R)$ carry a well-studied $\mathbb{C}\mathfrak{S}_n$-module structure for any $R \subseteq [n-1]$. As a consequence, our results recover and extend several known γ-positivity phenomena in symmetric functions and permutation enumeration.

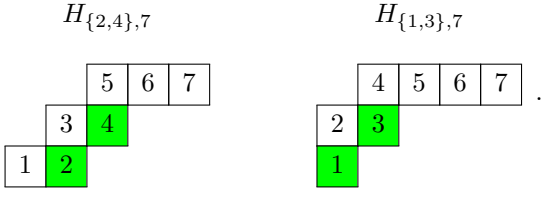
For background on representations of $\mathfrak{S}_n$, standard Young tableaux, and symmetric functions, we refer to Wachs [39] and Stanley [33]. For permutation statistics such as exc, inv, and maj, see Chapter 1 of Stanley [34].

Let ch denote the *Frobenius characteristic map*, which assigns to each isomorphism class of $\mathbb{C}\mathfrak{S}_n$-modules a symmetric function of degree n. Let ps_q denote the *stable principal specialization*, defined by substituting $x_i = q^{i-1}$ in a symmetric function. In place of the equivariant Hilbert series, it is often convenient to use the graded Frobenius series to encode a graded $\mathfrak{S}_n$-representation as a polynomial with coefficients in symmetric functions. For a graded $\mathbb{C}\mathfrak{S}_n$-module $V = \bigoplus_i V_i$, the *graded Frobenius series* of V is defined by

$$\mathsf{grFrob}(V, t) \coloneqq \sum_i \mathrm{ch}(V_i)\, t^i.$$

Let λ, μ be two partitions such that $\mu \subseteq \lambda$ (i.e. $\mu_i \leq \lambda_i$ for all i). The connected skew shape λ/μ is said to be a *ribbon* (or a *skew hook* or a *border strip*) if two consecutive rows always overlap in exactly one cell. A ribbon $H_{R,n}$ can be described in terms of the number n of its cells and its *descent set* $R \subseteq [n-1]$. Given a ribbon of n cells, we label its cells with numbers 1 through n, starting from the southwestern end cell, through the adjacent cell, ending at the northeastern end cell. We say a cell i is a *descent* of the ribbon if cell $i+1$ is directly above cell i. Then the collection of all its descents is called its *descent set*.

Example 4.1. Let $n = 7$. Then the diagrams of $H_{\{2,4\},7}$ and $H_{\{1,3\},7}$ are



Let $S^{\lambda/\mu}$ be the *Specht module* of skew shape λ/μ. If λ/μ is a ribbon $H_{R,n}$, the corresponding Specht module $S^{H_{R,n}}$ is known as a *Foulkes representation*. We summarize some facts of this representation from [32, Theorem 4.3] and [39, Theorem 3.4.4] in Theorem 4.2 (Note that Theorem 4.2 (i) was first proved by Solomon [30] in a different expression for any finite Coxeter group).

For a partition λ of n, denoted by $\lambda \vdash n$, let $SYT(\lambda)$ be the set of standard Young tableaux of shape λ. For $P \in SYT(\lambda)$, let $\mathsf{DES}(P)$ be the set of entries (called descents) $i \in [n-1]$ such that $i+1$ appears in a lower row than i in P, and $\mathsf{des}(P) = |\mathsf{DES}(P)|$.

Theorem 4.2. *Let $R \subseteq [n-1]$.*

(i) *Then*
$$\tilde{H}_{|R|-1}((B_n)_R) \cong_{\mathfrak{S}_n} S^{H_{R,n}} \cong_{\mathfrak{S}_n} \bigoplus_{\lambda \vdash n} b_\lambda(R) S^\lambda,$$
where $b_\lambda(R)$ is the number of $P \in SYT(\lambda)$ for which $\mathsf{DES}(P) = R$.

(ii) *Let $\mathbb{P}$ denote the set of positive integers and $\mathbb{P}_n$ the set of words of length n over $\mathbb{P}$. For $w = w_1 \dots w_n \in \mathbb{P}_n$, we say $i \in \mathsf{DES}(w)$ if $w_i > w_{i+1}$ and write $\mathbf{x}_w = x_{w_1} \dots x_{w_n}$. Then the Frobenius characteristic is the ribbon Schur function*
$$\mathrm{ch}(\tilde{H}_{|R|-1}((B_n)_R)) = s_{H_{R,n}} = \sum_{\substack{w \in \mathbb{P}_n \\ \mathsf{DES}(w)=R}} \mathbf{x}_w,$$

Applying Theorem 3.5 to the uniform matroids $U_{r,n}$ together with Theorem 2.1, one obtains the graded Frobenius series of $A(U_{r,n})_{\mathbb{C}}$. Applying the operator $\prod_{i=1}^{n}(1-q^i)\mathrm{ps}_q(-)$ to the graded Frobenius series gives the Hilbert series of q-Uniform matroid $U_{r,n}(q)$ (See Section 3 in [24] for the explanation). Note that the stable principal specialization on the ribbon Schur function gives
$$\prod_{i=1}^{n}(1-q^i)\mathrm{ps}_q(s_{H_{R,n}}) = \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \mathsf{DES}(\sigma)=R}} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \mathsf{DES}(\sigma)=R}} q^{\mathsf{inv}(\sigma)}$$
(see [29, Thm. 4.4] and its proof for a detailed discussion of the above identity). We obtain the following corollary, which generalizes Corollary 4.10 and 4.11 of [24]

Corollary 4.3. *For any positive integer n and $1 \le r \le n$,*
$$\begin{aligned}\mathsf{grFrob}(A(U_{r,n})_{\mathbb{C}}, t) &= \sum_{R \in \mathrm{Stab}([2,r-1])} s_{H_{R,n}} t^{|R|}(1+t)^{r-1-2|R|} \\ &= \sum_{k=0}^{\lfloor \frac{r-1}{2} \rfloor} \left(\sum_{\substack{R \in \mathrm{Stab}([2,r-1]) \\ |R|=k}} s_{H_{R,n}} \right) t^k (1+t)^{r-1-2k}\end{aligned}$$
and
$$\mathsf{Hilb}(A(U_{r,n}(q)), t) = \sum_{k=0}^{\lfloor \frac{r-1}{2} \rfloor} \xi_{r,n,k}(q) t^k (1+t)^{r-1-2k}$$
where
$$\xi_{r,n,k}(q) = \sum_{\sigma} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\sigma} q^{\mathsf{inv}(\sigma)}$$
and the sum runs through all $\sigma \in \mathfrak{S}_n$ for which $\mathsf{DES}(\sigma) \in \mathrm{Stab}([2, r-1])$ has k elements.

Applying Theorem 3.9 and the stable principal specialization as in Corollary 4.3, we obtain the augmented counterpart of Corollary 4.3, which generalizes Corollary 4.13 and 4.14 of [24].

Corollary 4.4. *For any positive integer n and $1 \le r \le n$,*
$$\begin{aligned}\mathsf{grFrob}(\widetilde{A}(U_{r,n})_{\mathbb{C}}, t) &= \sum_{R \in \mathrm{Stab}([r-1])} s_{H_{R,n}} t^{|R|}(1+t)^{r-2|R|} \\ &= \sum_{k=0}^{\lfloor \frac{r}{2} \rfloor} \left(\sum_{\substack{R \in \mathrm{Stab}([r-1]) \\ |R|=k}} s_{H_{R,n}} \right) t^k (1+t)^{r-2k}\end{aligned}$$

and

$$\mathsf{Hilb}(\widetilde{A}(U_{r,n}(q)),t)=\sum_{k=0}^{\lfloor \frac{r}{2} \rfloor}\widetilde{\xi}_{r,n,k}(q)t^k(1+t)^{r-2k},$$

where

$$\widetilde{\xi}_{r,n,k}(q)=\sum_{\sigma} q^{\mathsf{maj}(\sigma^{-1})}=\sum_{\sigma} q^{\mathsf{inv}(\sigma)}$$

and the sum runs through all $\sigma\in\mathfrak{S}_n$ *for which* $\mathsf{DES}(\sigma)\in \mathrm{Stab}([r-1])$ *has* k *elements.*

Recall from Corollary 4.2 and 4.4 of [24] that

$$\mathsf{grFrob}(A(U_{n-1,n}),t)=t^{-1}Q_n^0(\mathbf{x},t), \tag{4}$$

$$\mathsf{grFrob}(A(U_{n,n}),t)=\mathsf{grFrob}(\widetilde{A}(U_{n-1,n}),t)=Q_n(\mathbf{x},t), \tag{5}$$

$$\mathsf{grFrob}(\widetilde{A}(U_{n,n}),t)=\widetilde{Q}_n(\mathbf{x},t). \tag{6}$$

Here Q_n^0, Q_n, $\widetilde{Q}_n$ are the *Eulerian quasisymmetric functions* introduced by Shareshian and Wachs [28, 29]. They admit combinatorial interpretations as enumerators over the sets of derangements D_n (permutations with no fixed points), permutations $\mathfrak{S}_n$, and *decorated permutations* $\widetilde{\mathfrak{S}}_n$ (permutations with two types of fixed points), respectively, satisfying

$$D_n\subset\mathfrak{S}_n\subset\widetilde{\mathfrak{S}}_n.$$

More precisely,

$$Q_n^0(\mathbf{x},t)=\sum_{\sigma\in D_n}F_{\mathsf{DEX}(\sigma),n}(\mathbf{x})t^{\mathsf{exc}(\sigma)},\quad Q_n(\mathbf{x},t)=\sum_{\sigma\in\mathfrak{S}_n}F_{\mathsf{DEX}(\sigma),n}(\mathbf{x})t^{\mathsf{exc}(\sigma)}, \tag{7}$$

and

$$\widetilde{Q}_n(\mathbf{x},t)=\sum_{\sigma\in\widetilde{\mathfrak{S}}_n}F_{\mathsf{DEX}(\sigma),n}(\mathbf{x})t^{\mathsf{exc}(\sigma)+1}. \tag{8}$$

Here $F_{S,n}(\mathbf{x})$ for $S\subseteq[n-1]$ denotes Gessel's *fundamental quasisymmetric function* [1]

$$F_{S,n}(\mathbf{x}):=\sum_{\substack{i_1\geq i_2\geq\cdots\geq i_n\\ j\in S\Rightarrow i_j>i_{j+1}}}x_{i_1}x_{i_2}\dots x_{i_n}\quad\text{with } F_{\phi,0}:=1.$$

See [24] for the definition of the descent-like statistic DEX on $\widetilde{\mathfrak{S}}_n$ and further details on these symmetric functions. Finally, by Lemma 6.20 of [23], applying the operator $\prod_{i=1}^n(1-q^i)\mathrm{ps}_q(-)$ to Q_n^0, Q_n, and $\widetilde{Q}_n$ yields the q-analogs of Eulerian type polynomials:

$$d_n(q,t)=\sum_{\sigma\in D_n}q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)}t^{\mathsf{exc}(\sigma)},\quad A_n(q,t)=\sum_{\sigma\in\mathfrak{S}_n}q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)}t^{\mathsf{exc}(\sigma)},$$

and

$$\widetilde{A}_n(q,t):=1+\sum_{k=1}^n\begin{bmatrix}n\\k\end{bmatrix}_q A_k(q,t)=\sum_{\sigma\in\widetilde{\mathfrak{S}}_n}q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)}t^{\mathsf{exc}(\sigma)+1}.$$

Let NDD_n be the set of words $w\in\mathbb{P}_n$ with no double descents, i.e. $\mathsf{DES}(w)\in\mathrm{Stab}([n-1])$. Combining (4) and Corollary 4.3, we recover the following collection of results from [28, Theorem 7.3, Equation (7.9)] and [29, Theorem 6.1].

Corollary 4.5. *For positive integer* $n\geq 2$*, we have*

$$\begin{aligned}Q_n^0(\mathbf{x},t)&=\sum_{R\in\mathrm{Stab}([2,n-2])}s_{H_{R,n}}t^{|R|}(1+t)^{n-1-2|R|}\\&=\sum_{\substack{w\in NDD_n\\ w_1\leq w_2,\ w_{n-1}\leq w_n}}\mathbf{x}_w t^{\mathsf{des}(w)+1}(1+t)^{n-2-2\mathsf{des}(w)}\\&=\sum_{k=0}^{\lfloor\frac{n-2}{2}\rfloor}\xi_{n,k}^0(\mathbf{x})t^{k+1}(1+t)^{n-2-2k},\end{aligned}$$

[1] Note that our definition of the fundamental quasisymmetric function is the reverse of the usual one. The usual definition is summed over $i_1\leq i_2\leq\cdots\leq i_n$ with $j\in S$ implying $i_j<i_{j+1}$.

where

$$\xi^0_{n,k}(\mathbf{x}) = \sum_{\substack{R\in \mathrm{Stab}([2,n-2])\\ |R|=k}} s_{H_{R,n}} = \sum_{\substack{w\in NDD_n\\ w_1\le w_2,\ w_{n-1}\le w_n\\ \mathsf{des}(w)=k}} \mathbf{x}_w.$$

And

$$d_n(q,t) = \sum_{k=0}^{\lfloor \frac{n-2}{2}\rfloor} \xi^0_{n,k}(q) t^{k+1}(1+t)^{n-2-2k},$$

where

$$\xi^0_{n,k}(q) = \sum_{\sigma} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\sigma} q^{\mathsf{inv}(\sigma)}$$

and both of the sums run through $\sigma \in D_n$ which has k descents and no double descents, no first, no last descent.

Combining (5) with Corollary 4.3 and Corollary 4.4, we recover the following collection of results from [28, Theorem 7.3, Equation (7.8)] and [26, Equations (1.4) and (6.1)].

Corollary 4.6. *For positive integer $n \ge 1$, we have the following expressions*

$$\begin{aligned} Q_n(\mathbf{x},t) &= \sum_{R\in \mathrm{Stab}([2,n-1])} s_{H_{R,n}} t^{|R|}(1+t)^{n-1-2|R|} = \sum_{R\in \mathrm{Stab}([n-2])} s_{H_{R,n}} t^{|R|}(1+t)^{n-1-2|R|} \\ &= \sum_{\substack{w\in NDD_n\\ w_1\le w_2}} \mathbf{x}_w t^{\mathsf{des}(w)}(1+t)^{n-1-2\mathsf{des}(w)} = \sum_{\substack{w\in NDD_n\\ w_{n-1}\le w_n}} \mathbf{x}_w t^{\mathsf{des}(w)}(1+t)^{n-1-2\mathsf{des}(w)} \\ &= \sum_{k=0}^{\lfloor \frac{n-1}{2}\rfloor} \xi_{n,k}(\mathbf{x}) t^k (1+t)^{n-1-2k}, \end{aligned}$$

where

$$\begin{aligned} \xi_{n,k}(\mathbf{x}) &= \sum_{\substack{R\in \mathrm{Stab}([2,n-1])\\ |R|=k}} s_{H_{R,n}} = \sum_{\substack{R\in \mathrm{Stab}([n-2])\\ |R|=k}} s_{H_{R,n}} \\ &= \sum_{\substack{w\in NDD_n\\ w_1\le w_2\\ \mathsf{des}(w)=k}} \mathbf{x}_w = \sum_{\substack{w\in NDD_n\\ w_{n-1}\le w_n\\ \mathsf{des}(w)=k}} \mathbf{x}_w. \end{aligned}$$

And

$$A_n(q,t) = \sum_{k=0}^{\lfloor \frac{n-1}{2}\rfloor} \xi_{n,k}(q) t^{k+1}(1+t)^{n-1-2k},$$

where

$$\xi_{n,k}(q) = \sum_{\sigma} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\sigma} q^{\mathsf{inv}(\sigma)}$$

and both of the sums run through either:

- *$\sigma \in \mathfrak{S}_n$ with k descents and no double descents, no first descent.*
- *$\sigma \in \mathfrak{S}_n$ with k descents and no double descents, no last descent.*

Remark 4.1. In [3, Theorem 2.1], Athanasiadis summarizes different interpretations of the γ-coefficients for the Eulerian polynomial. In particular, the interpretations are given by counting permutations in $\mathfrak{S}_n$ related to the descent set in $\mathrm{Stab}([2,n-1])$ or in $\mathrm{Stab}([n-2])$. This fact is lifted to the q-analog and the symmetric function analog in Corollary 4.6. The agreement of the two interpretations regarding $\mathrm{Stab}([2,n-1])$ and $\mathrm{Stab}([n-2])$ was shown algebraically by (5).

Combining (6) and Corollary 4.4, we recover gives the following collection of results of [29, Theorems 3.4 and 4.5]

Corollary 4.7 (cf. Shareshian and Wachs [29], Athanasiadis [3]). *For positive integer $n \geq 1$, we have*

$$\begin{aligned}\widetilde{Q}_n(\mathbf{x},t) &= \sum_{R\in \mathrm{Stab}([n-1])} s_{H_{R,n}} t^{|R|}(1+t)^{n-2|R|} \\ &= \sum_{w\in NDD_n} \mathbf{x}_w t^{\mathsf{des}(w)}(1+t)^{n-2\mathsf{des}(w)} \\ &= \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \widetilde{\xi}_{n,k}(\mathbf{x}) t^k (1+t)^{n-2k},\end{aligned}$$

where

$$\widetilde{\xi}_{n,k}(\mathbf{x}) = \sum_{\substack{R\in \mathrm{Stab}([n-1]) \\ |R|=k}} s_{H_{R,n}} = \sum_{\substack{w\in NDD_n \\ \mathsf{des}(w)=k}} \mathbf{x}_w.$$

And

$$\widetilde{A}_n(q,t) = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \widetilde{\xi}_{n,k}(q) t^k (1+t)^{n-2k},$$

where

$$\widetilde{\xi}_{n,k}(q) = \sum_{\sigma} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\sigma} q^{\mathsf{inv}(\sigma)}$$

and both of the sums run through $\sigma \in \mathfrak{S}_n$ with k descents and no double descents.

5. A problem proposed by Athanasiadis

In this section, we apply some results from Section 4 to address a problem proposed by Athanasiadis [3, Problem 2.5].

For $\sigma \in \mathfrak{S}_n$, we set

$$\mathsf{des}^*(\sigma) = \begin{cases} \mathsf{des}(\sigma) & \text{if } \sigma_1 = 1 \\ \mathsf{des}(\sigma) - 1 & \text{if } \sigma_1 > 1 \end{cases}.$$

Recall the following (p,q)-analog of the γ-expansion of the Eulerian polynomial in [3, Theorem 2.3].

Theorem 5.1. *For all positive integer $n \geq 1$,*

$$\sum_{\sigma\in\mathfrak{S}_n} p^{\mathsf{des}^*(\sigma)} q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)} t^{\mathsf{exc}(\sigma)} = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \xi_{n,k}(p,q) t^k (1+t)^{n-1-2k}$$

where

$$\xi_{n,k}(p,q) = \sum_{\sigma} p^{\mathsf{des}(\sigma^{-1})} q^{\mathsf{maj}(\sigma^{-1})}$$

summing over permutations $\sigma \in \mathfrak{S}_n$ with k descents and no double descents, no last descent.

Recall a similar q-analog of the γ-expansion of the binomial Eulerian polynomial $\widetilde{A}_n(t)$ found by Shareshian and Wachs [29] is given in Corollary 4.7. Athanasiadis proposed the following problem.

Problem 5.2 ([3, Problem 2.5]). *Find a p-analog of the q-polynomial in Corollary 4.7 that is similar to Theorem 5.1.*

The idea of finding such a (p,q)-analog of $\widetilde{A}_n(t)$ is similar to the proof of Theorem 5.1. But the key step is the $\mathfrak{S}_n$-equivariant enumerator of decorated permutations in (8)[2].

Before answering Athanasiadis' question, let us first generalize Corollary 2.41 in [3], which states the γ-positivity for the coefficient of Schur-expansions of $Q_n^0(\mathbf{x},t)$ and $Q_n(\mathbf{x},t)$. Note that Q_n and Q_n^0 are related to the graded Frobenius series of the Chow ring of $U_{n,n}$ and $U_{n-1,n}$ respectively. Here we generalize these results to all uniform matroids $U_{r,n}$ and their augmented counterparts.

Let

$$\mathsf{grFrob}(A(U_{r,n})_{\mathbb{C}},t) = \sum_{\lambda\vdash n} P_\lambda^r(t) s_\lambda \text{ and } \mathsf{grFrob}(\widetilde{A}(U_{r,n})_{\mathbb{C}},t) = \sum_{\lambda\vdash n} \widetilde{P_\lambda^r}(t) s_\lambda.$$

[2]This interpretation is proved in Liao [23, Theorem 6.21]. The original definition given by Shareshian and Wachs [29] is $\widetilde{Q}_n(\mathbf{x},t) := h_n + t\sum_{k=1}^n h_{n-k} Q_k(\mathbf{x},t)$.

Corollary 5.3. *For $\lambda \vdash n$, we have*

$$P_\lambda^r(t) = \sum_{\substack{P\in SYT(\lambda)\\ \mathsf{DES}(P)\in \mathrm{Stab}([2,r-1])}} t^{\mathsf{des}(P)}(1+t)^{r-1-2\mathsf{des}(P)} = \sum_{k=0}^{\lfloor \frac{r-1}{2}\rfloor} \xi_{r,\lambda,k}\ t^k(1+t)^{r-1-2k}$$

and

$$\widetilde{P}_\lambda^r(t) = \sum_{\substack{P\in SYT(\lambda)\\ \mathsf{DES}(P)\in \mathrm{Stab}([r-1])}} t^{\mathsf{des}(P)}(1+t)^{r-2\mathsf{des}(P)} = \sum_{k=0}^{\lfloor \frac{r}{2}\rfloor} \widetilde{\xi}_{r,\lambda,k}\ t^k(1+t)^{r-2k}$$

where $\xi_{r,\lambda,k}$ (respectively, $\widetilde{\xi}_{r,\lambda,k}$) is the number of tableaux $P \in SYT(\lambda)$ for which $\mathsf{DES}(P) \in \mathrm{Stab}([2, r-1])$ (respectively, $\mathrm{Stab}([r-1])$) has k elements. In particular, $P_\lambda^r(t)$ and $\widetilde{P}_\lambda^r(t)$ are palindromic and unimodal for all λ and r.

Proof. Consider the non-augmented case. Apply Corollary 4.3 and use Theorem 2.1 (i) to expand $s_{H_{R,n}}$ into Schur functions of regular shapes. We have

$$\mathsf{grFrob}(A(U_{r,n})_{\mathbb{C}}, t) = \sum_{\lambda\vdash n}\left(\sum_{R\in \mathrm{Stab}([2,r-1])} b_\lambda(R)t^{|R|}(1+t)^{n-2|R|}\right)s_\lambda(\mathbf{x}).$$

Comparing the coefficient of $s_\lambda(\mathbf{x})$, we obtain

$$P_\lambda^r(t) = \sum_{R\in \mathrm{Stab}([2,r-1])} b_\lambda(R)t^{|R|}(1+t)^{r-1-2|R|} = \sum_{\substack{P\in SYT(\lambda)\\ \mathsf{DES}(P)\in \mathrm{Stab}([2,r-1])}} t^{\mathsf{des}(P)}(1+t)^{r-1-2\mathsf{des}(P)}.$$

The augmented can be worked out similarly. □

Remark 5.1. For general $1 \le r \le n$, the symmetric function polynomials $\mathsf{grFrob}(A(U_{r,n})_{\mathbb{C}}, t)$ and $\mathsf{grFrob}(\widetilde{A}(U_{r,n})_{\mathbb{C}}, t)$ each has two difference expressions as shown in [24, Theorem 4.1 and Theorem 4.3]. Unfortunately, we do not have nice combinatorial expressions as the cases $r = n-1,\ n$ in (7) and (8).

In particular, when $r = n$ in the augmented case of Corollary 5.3, it states that the Schur-expansion of the binomial Eulerian quasisymmetric function $\widetilde{Q}_n(\mathbf{x}, t)$ has γ-positive coefficients. This will be used in the proof of Theorem 5.4.

We used stable principal specialization to obtain many interesting q-analogs in the previous sections. To obtain a (p, q)-analog of $\widetilde{A}_n(t)$, we need a different specialization. Given a formal power series $f(\mathbf{x}) = f(x_1, x_2, \ldots)$ and a positive integer $m \ge 1$, the *principal specialization* $\mathrm{ps}_{q,m}(f)$ of f is obtain from f by setting

$$x_i = \begin{cases} q^{i-1} & \text{if } 1 \le i \le m \\ 0 & \text{if } i \ge m+1 \end{cases}.$$

Then the principal specialization of $F_{S,n}$ (see [17, Lemma 5.2]) and $s_\lambda(\mathbf{x})$ (see [33, Proposition 7.19.12]) are given by the formulas

$$\sum_{m\ge 1} \mathrm{ps}_{q,m}(F_{S,n}(\mathbf{x}))p^{m-1} = \frac{p^{|S|}q^{\sum_{i\in S} i}}{(1-p)(1-pq)\ldots(1-pq^n)} \tag{9}$$

and

$$\sum_{m\ge 1} \mathrm{ps}_{q,m}(s_\lambda(\mathbf{x}))p^{m-1} = \frac{\sum_{P\in SYT(\lambda)} p^{\mathsf{des}(P)}q^{\mathsf{maj}(P)}}{(1-p)(1-pq)\ldots(1-pq^n)}. \tag{10}$$

For any decorated permutation $\sigma \in \widetilde{\mathfrak{S}}_n$, we set

$$\widetilde{\mathsf{des}}(\sigma) = \begin{cases} \mathsf{des}(\sigma) & \text{if } \sigma_1 = 0 \text{ or } 1 \\ \mathsf{des}(\sigma) - 1 & \text{if } \sigma_1 > 1 \end{cases}.$$

Now we are ready to state our (p, q)-analog for $\widetilde{A}_n(t)$.

Theorem 5.4. *For a positive integer $n \geq 1$,*

$$\sum_{\sigma\in\widetilde{\mathfrak{S}}_n} p^{\widetilde{\mathsf{des}}(\sigma)} q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)} t^{\mathsf{exc}(\sigma)+1} = \sum_{\substack{\sigma\in\mathfrak{S}_n \\ \mathsf{DES}(\sigma)\in\mathrm{Stab}([n-1])}} p^{\mathsf{des}(\sigma^{-1})} q^{\mathsf{maj}(\sigma^{-1})} t^{\mathsf{des}(\sigma)} (1+t)^{n-2\mathsf{des}(\sigma)}$$
$$= \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \widetilde{\xi}_{n,k}(p,q) t^k (1+t)^{n-2k}$$

where

$$\widetilde{\xi}_{n,k}(p,q) = \sum_{\sigma} p^{\mathsf{des}(\sigma^{-1})} q^{\mathsf{maj}(\sigma^{-1})}$$

and the sum runs through $\sigma \in \mathfrak{S}_n$ with k descents and no double descents.

Proof. First applying principal specialization on (8) and by (9), we have

$$\sum_{m\geq 1} \mathrm{ps}_{q,m}(\widetilde{Q}_n(\mathbf{x},t)) p^{m-1}$$
$$= \sum_{\sigma\in\widetilde{\mathfrak{S}}_n} \sum_{m\geq 1} \mathrm{ps}_{q,m}(F_{\mathsf{DEX}(\sigma),n}(\mathbf{x})) p^{m-1} t^{\mathsf{exc}(\sigma)+1}$$
$$= \frac{\sum_{\sigma\in\widetilde{\mathfrak{S}}_n} p^{\widetilde{\mathsf{des}}(\sigma)} q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)} t^{\mathsf{exc}(\sigma)+1}}{(1-p)(1-pq)\dots(1-pq^n)} \quad \text{(by Lemma 6.20 in [23])}.$$

On the other hand,

$$\sum_{m\geq 1} \mathrm{ps}_{q,m}(\widetilde{Q}_n(\mathbf{x},t)) p^{m-1}$$
$$= \sum_{\lambda\vdash n} \widetilde{P}^n_\lambda(t) \sum_{m\geq 1} \mathrm{ps}_{q,m}(s_\lambda(\mathbf{x})) p^{m-1}$$
$$= \frac{\sum_{\lambda\vdash n} \widetilde{P}^n_\lambda(t) \sum_{P\in SYT(\lambda)} p^{\mathsf{des}(P)} q^{\mathsf{maj}(P)}}{(1-p)(1-pq)\dots(1-pq^n)} \quad \text{(by (10))}.$$

Therefore, we have

$$\sum_{\sigma\in\widetilde{\mathfrak{S}}_n} p^{\widetilde{\mathsf{des}}(\sigma)} q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)} t^{\mathsf{exc}(\sigma)+1} = \sum_{\lambda\vdash n} \sum_{P\in SYT(\lambda)} \widetilde{P}^n_\lambda(t) p^{\mathsf{des}(P)} q^{\mathsf{maj}(P)}$$
$$= \sum_{\lambda\vdash n} \sum_{P\in SYT(\lambda)} \sum_{\substack{Q\in SYT(\lambda) \\ \mathsf{DES}(Q)\in\mathrm{Stab}([n-1])}} p^{\mathsf{des}(P)} q^{\mathsf{maj}(P)} t^{\mathsf{des}(Q)} (1+t)^{n-2\mathsf{des}(Q)} \quad \text{(by Corollary 5.3)}$$
$$= \sum_{\substack{\sigma\in\mathfrak{S}_n \\ \mathsf{DES}(\sigma)\in\mathrm{Stab}([n-1])}} p^{\mathsf{des}(\sigma^{-1})} q^{\mathsf{maj}(\sigma^{-1})} t^{\mathsf{des}(\sigma)} (1+t)^{n-2\mathsf{des}(\sigma)}.$$

The last equality is obtained by applying the RSK algorithm described in Section 7.11 of Stanley [33, Lemma 7.23.1]. □

6. A remark on braid maroids

The *Braid matroids* K_n are the matroid associated with the braid arrangement $\{x_i - x_j = 0\}_{1\leq i<j\leq n}$, and equivalently, the graphical matroid of the complete graph of n vertices. Its lattice of flats is the intersection lattice of the braid arrangement, and is given by the partition lattice Π_n. In [12], Ferroni, Matherne, Stevens, and Vecchi proposed a problem of studying the Hilbert series of the Chow and augmented Chow ring of K_n. Applying Theorem 3.5 and Theorem 3.9, we have

$$\mathsf{Hilb}_G(A(K_n),t) = \sum_{S\in\mathrm{Stab}([2,n-2])} [\tilde{H}_{|S|-1}((\Pi_n)_S)] t^{|S|} (1+t)^{n-2-2|S|}$$

and

$$\mathsf{Hilb}_G(\widetilde{A}(K_n),t) = \sum_{S\in\mathrm{Stab}([n-2])} [\tilde{H}_{|S|-1}((\Pi_n)_S)] t^{|S|} (1+t)^{n-1-2|S|}.$$

These give an explicit connection between the (augmented) Chow ring of K_n and the top reduced homology of the rank-selected partition lattice $(\Pi_n)_S$. The study of $\mathfrak{S}_n$-representation $\tilde{H}_{|S|-1}((\Pi_n)_S)$

is initiated from Stanley's classical paper [32], and has been studied in depth since then. A nice survey for this is Sundaram [38]. However, the $\mathfrak{S}_n$-module structures of $\tilde{H}_{|S|-1}((\Pi_n)_S)$ seem to vary a lot, and there is no unified characterization for every subset S so far. Regarding the dimension (rank) of $\tilde{H}_{|S|-1}((\Pi_n)_S)$, using the theory of EL-labeling, there is a general formula for all subsets S, see Sundaram [37, Prop. 2.18] or Björner [5, p.276, Exercise 7.33].

7. New developments

Since the preliminary version of this paper was posted on `arXiv`, there have been many interesting developments.

7.1. **Chow polynomials for posets.** Ferroni, Matherne, and Vecchi [13] generalized the notion of Chow polynomials from matroids to locally finite, weakly ranked posets, placing them within the framework of Kazhdan–Lusztig–Stanley theory for posets.

Given such a poset P and a choice of a distinguished element κ in the incidence algebra of P, called a *P-kernel*, they define the corresponding *κ-Chow polynomial of P*. For suitable choices of κ, this construction recovers and extends Chow-type invariants arising in the study of Eulerian posets and the Kazhdan–Lusztig theory of Coxeter groups, and allows for a systematic investigation of their positivity phenomena.

In particular, when κ is the characteristic polynomial of P and P is a lattice of flats of a matroid M, one recovers the Chow polynomial of M. As noted in Remark 3.3, using an approach similar to ours, they proved that if P is Cohen–Macaulay, then both the characteristic Chow and augmented Chow polynomials are γ-positive [13, Theorem 4.25]. We may extend Theorems 3.5 and 3.9 to Cohen-Macaulay posets as well.

Definition 7.1. Let P be a finite graded bounded poset of rank r, and G be a group of automorphisms of P. Define the *(G-)equivariant Chow polynomial* and *(G-)equivariant augmented Chow polynomial* of P [3], respectively, to be the following polynomials with coefficients in $R_{\mathbb{C}}(G)$:

$$\mathrm{H}_P(t) \coloneqq \sum_{S\subseteq[r-1]} \phi_{S,r}(t)[\alpha_P(S)]$$

and

$$^{\mathrm{aug}}\mathrm{H}_P(t) \coloneqq \sum_{S\subseteq[r-1]} \psi_{S,r}(t)[\alpha_P(S)],$$

where $\phi_{S,r}(t)$ and $\psi_{S,r}(t)$ are as in Proposition 3.4.

When G is trivial (or by taking dimensions of $\alpha_P(S)$), following from [13, Theorem 4.1] and its left augmented counterpart, the above polynomials specialize to the non-equivariant Chow and augmented Chow polynomials of P. The same arguments as in Theorems 3.5 and 3.9 then yield the following extension.

Theorem 7.2. *Let P be a finite bounded graded poset of rank r. Then:*

(i) *the equivariant Chow polynomial of P admits the γ-expansion*

$$\mathrm{H}_P(t) = \sum_{S\in\mathrm{Stab}([2,r-1])} [\beta_P(S)]t^{|S|}(1+t)^{r-1-2|S|}, \text{ and}$$

(ii) *the equivariant augmented Chow polynomial of P admits γ-expansion*

$$^{\mathrm{aug}}\mathrm{H}_P(t) = \sum_{S\in\mathrm{Stab}([r-1])} [\beta_P(S)]t^{|S|}(1+t)^{r-2|S|}.$$

In particular, if P is a Cohen-Macaulay poset, then $\mathrm{H}_P(t)$ and $^{\mathrm{aug}}\mathrm{H}_P(t)$ are equivariant γ-positive.

Along these lines, Ferroni, Matherne, and Vecchi further extend Conjecture 1.1 to this borader setting.

Conjecture 7.3 (Conjecture 4.26 in [13])**.** *Let P be a Cohen–Macaulay poset. The characteristic Chow polynomial of P is real-rooted.*

Note that the real-rootedness of the characteristic (left) augmented Chow polynomial of a Cohen–Macaulay poset would follow from [13, Corollary 4.6] together with the above conjecture.

[3] Recursive definitions of $\mathrm{H}_P(t)$ and $^{\mathrm{aug}}\mathrm{H}_P(t)$ analogous to the non-equivariant construction in [12] are given by Matherne and Nepel (personal communication).

7.2. **Uniform matroids $U_{r,n}$ and their generalizations.** The coefficients of the Chow and augmented Chow polynomials of the Boolean matroid $U_{n,n}$ are counted, respectively, by the numbers of loopless Schubert matroids and Schubert matroids of the corresponding ranks. This follows from the bijection described in [10, proof of Theorem 7.7] between the Feichtner–Yuzvinsky basis $\widetilde{\mathrm{FY}}(U_{n,n})$ and Schubert matroids. Hoster [19] extended this combinatorial interpretation to Chow and augmented Chow polynomials of uniform matroids $U_{r,n}$ for all r.

Brändén and Vecchi [8] proved that the Chow polynomials of uniform matroids $U_{r,n}$ are real-rooted, and also provided a new proof of the real-rootedness of the augmented Chow polynomials of $U_{r,n}$, independent of the earlier argument in [12, Theorem 1.10].

Hoster and Stump [20] used the non-equivariant analogs of Theorems 3.5 and 3.9 to prove real-rootedness of Chow and augmented Chow polynomials for posets $\widehat{P} = P \cup \{\hat{1}\}$ where P is a simplicial poset with positive h-vector. In particular, their argument yields yet another proof of real-rootedness for uniform matroids.

7.3. **Braid matroids K_n.** Kannan and Kühne [21] obtained a closed formula for the generating function of the $\mathfrak{S}_n$-equivariant Chow polynomial $\mathrm{H}_{K_n}(t)$ of the braid matroid K_n, expressed in terms of the $\mathfrak{S}_n$-equivariant compactly supported Poincaré polynomial of $\mathcal{M}_{0,n+1}$. They further showed that the Chow polynomial of K_n agrees with the Hilbert series of the Chow ring of the moduli space of genus-zero relative stable maps to $\mathbb{P}^1$.

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